LSU | College of Art + Design

LA 4008 & LA 7051 | Ecological Robotics

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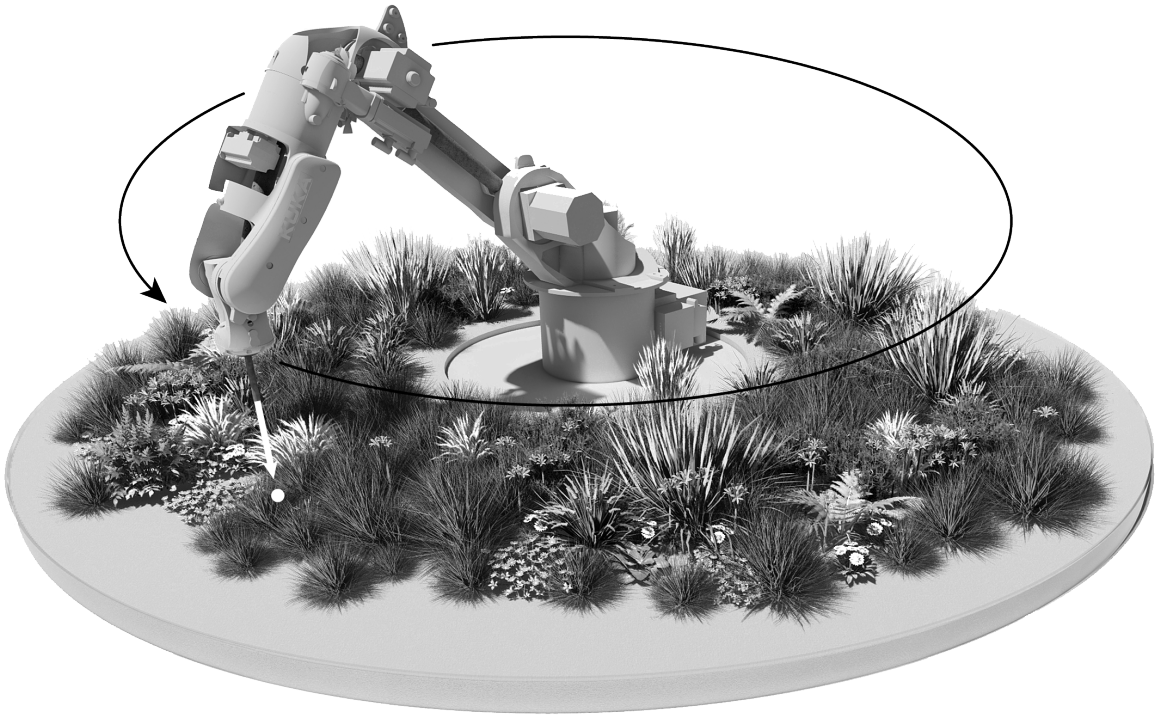
MWF 1:30pm–5:30pm

Course Description

This studio will explore ecological applications for robots. In this studio you will learn how to program industrial robots, design your own robotic tools, and robotically plant a landscape. Through a series of projects you will design a robotic process for algorithmic planting and test it both in the studio and in the field. The field site will be a set of plots at Burden where you will conduct a controlled planting experiment. As an introduction to robotics for designers, this studio will cover topics such as visual programming, generative design, robotic operations, 3D printing, 3D scanning, and drone photogrammetry.

Topics

Robotic Sandbox		Robotic Planting		Robotic Landscape	
1	Robotic Operations	6	3D Printing Seeds	11	Site Preparation
2	Robotic Programming	7	Laboratory Planting	12	Field Planting
3	Robotic Tools	8	Algorithmic Seeding	13	Drone Mapping
4	Visual Programming	9	Algorithmic Landforms	14	Documentation
5	Robotic Sandbox	10	3D Scanning	15	Exhibition